Flight Booking & Travel Data Analysis

- Analyzed trends for Revenue Performance.
- Analyzed trends for Travel class performance.
- Analyzed trends for pricing vs distance.
- Analyzed trends for Passenger Behavior.

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Introduction

This analysis provides a comprehensive overview of Flight Booking and Travel Data, focusing on key business and customer insights. It highlights overall performance metrics such as total bookings, revenue, distance traveled, flight duration, and passenger demographics.

The analysis further explores booking patterns across travel classes, ticket pricing trends by airline and departure city, distance versus ticket price relationships, and customer payment preferences.

Passenger Age was converted into custom age groups using calculated column in DAX, using **SWITCH Function:**

```
Age Group = SWITCH(TRUE(),flight_bookings_sample[Passenger_Age]<= 18,"0-18",flight_bookings_sample[Passenger_Age]<= 30,"19-30",flight_bookings_sample[Passenger_Age]<= 45,"31-45",flight_bookings_sample[Passenger_Age]<= 60,"60","60+")
```

1. Overall KPIs

- The dashboard records a total of **250** flight bookings, generating **$644.56K** in total revenue.

- Passengers have travelled a combined **1,889.24K kilometers**, indicating a strong presence of long-haul routes.

- The average flight duration is **11.04 hours**, reinforcing that many flights are **medium to long distance**.

- The average passenger age is **45.6 years**, suggesting that the majority of travelers are **working professionals or mid-career individuals**.

2. Booking Distribution by Travel Class

- **Premium Economy** leads with the **highest bookings (~0.07K)**, showing strong customer preference for **comfort at a moderate price**.
- **First Class, Business Class, and Economy** each contribute around **0.06K** bookings, indicating a **relatively balanced distribution**.

Insight: Customers are increasingly willing to pay extra for comfort without opting for luxury pricing, highlighting a shift toward value-added travel options.

3. Ticket Price by Airline

- **B Airlines** generates the highest ticket revenue **(~0.13M)**, making it the top-performing airline.
- **S Airlines and E Airlines** follow closely with around **0.11M** each, showing competitive pricing and demand.
- **L Airlines** and **Air I** contribute approximately **0.10M**, while **Q Airways** records the lowest revenue **(~0.09M)**.

Insight: Revenue concentration varies across airlines, indicating differences in pricing strategy, route network, or customer preference.

4.Ticket Price VS Distance by Travel Class

- **Premium Economy** covers the longest distance **(~524.46K km)** with ticket revenue of **~179.68K**, showing strong long-haul demand.

- **Business Class** generates the **highest ticket revenue (~488.97K)** despite covering a shorter distance **(~158.01K km)**, indicating higher ticket prices.

- **First Class** shows moderate distance **(~452.34K km)** and revenue **(~181.34K)**.

- **Economy Class** covers the least distance **(~423.47K km)** and generates the lowest revenue **(~125.54K)**.

Insight: Higher travel classes generate more revenue even over shorter distances due to premium pricing.

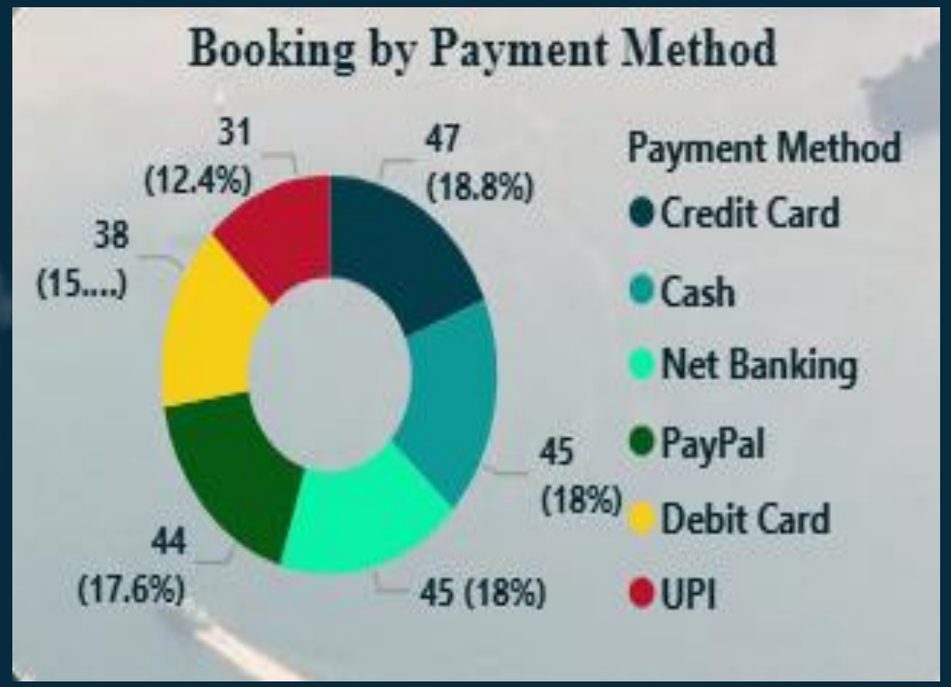
5.Ticket Price by Departure City

- **Mumbai** is the **highest revenue-generating** departure city **(~98K).**
- **London (~91K)** and **Delhi (~83K)** follow, highlighting strong **international** and **domestic travel demand.**
- **Singapore (~81K), Frankfurt (~78K)**, and **Paris (~76K)** show steady contributions.
- **Dubai (~73K)** and **New York (~65K)** contribute comparatively **lower ticket revenue.**

Insight: Major metro cities act as primary revenue hubs due to higher passenger traffic.

6. Booking By Payment Method

- **Credit Card** is the most preferred payment method, accounting for **47 bookings (18.8%)**, indicating strong reliance on card-based transactions.
- **Cash and Net Banking** show equal adoption, each contributing **45 bookings (~18%)**.
- **PayPal** follows closely with **44 bookings (17.6%)**, indicating strong digital wallet usage.
- **Debit Card** usage is moderate at **38 bookings (~15%)**, indicating consistent but lower preference compared to credit cards.
- **UPI** is the least used payment method, with **31 bookings (12.4%)**, suggesting potential scope for increasing adoption through offers or incentives.



Insight: Digital payment methods dominate bookings, with Credit Cards leading, while UPI shows the lowest adoption, indicating an opportunity to grow instant-payment usage.

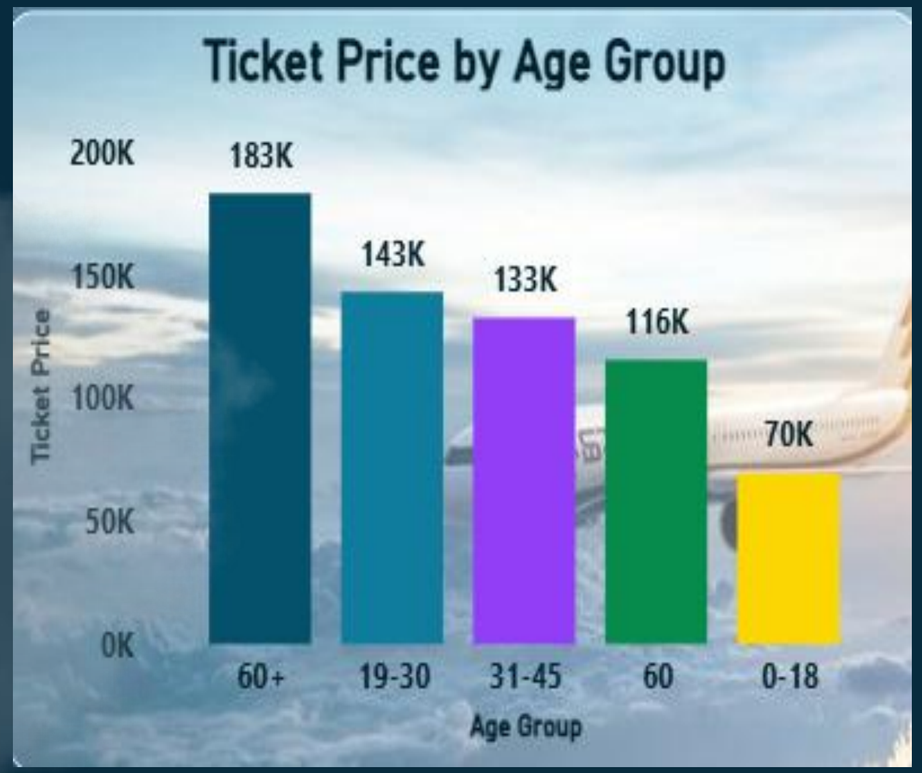
7. Ticket Price By Age Group

The **60+ age group** contributes the highest ticket value **(~183K)**, possibly due to premium travel preferences or longer routes.

The **19–30 age group** follows with **~143K**, reflecting frequent travel by young professionals.

The **31–45** group contributes **~133K**, showing steady but slightly lower spending.

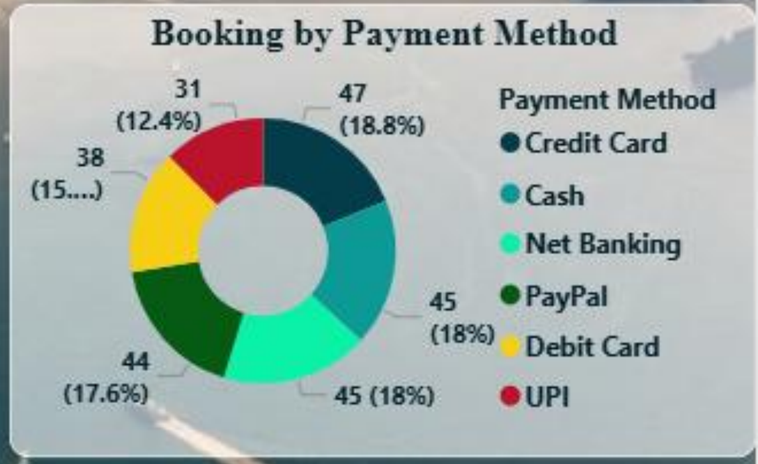
Passengers aged up to **60** contribute **~116K**, while the **0-18 age group has the lowest ticket value (~70K)**.



Insight: Ticket spending increases with age, peaking in Senior travelers.

Flight Booking & Travel Data Analysis

250	644.56K	1,889.24K	11.04	45.60
Total Booking	Total Revenue	Total Distance Travelled(km)	Average Flight Duration(hr)	Average Passenger Age



Strategic Recommendations

Short-Term Recommendations

- Focus promotional offers on Premium Economy, as it shows the highest booking demand.
- Strengthen partnerships with top-performing airlines to maximize revenue.
- Promote digital payment methods with small incentives to improve checkout convenience.

Mid-Term Recommendations

- Optimize pricing strategies by distance and travel class to improve revenue per booking.
- Launch targeted marketing campaigns for high-spending age groups (45+ and 60+).
- Improve route planning from high-revenue departure cities such as Mumbai and London.

Long-Term Recommendations

- Expand premium and business-class offerings on long-haul routes to increase profitability.
- Use passenger demographic trends to develop personalized travel packages.
- Implement advanced analytics for demand forecasting and dynamic pricing to forecast a) Booking by route b)Booking by travel class, c)Seasonal Peaks using ARIMA/SARIMA or MOVING AVERAGE.

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